Research Report on Baseline Swing Form Definition and Automated Diagnostic Logic Based on 2D Side-View Video Analysis in Youth Baseball

To construct an automated image analysis and scoring system for batting forms in youth baseball, it is essential to establish an algorithm that converts 2D skeletal keypoints (Pose Estimation) obtained from a single 2D side-view camera into comprehensive kinematic metrics and quantitatively evaluates deviations from an ideal kinematic sequence. This report defines baseline swing mechanics tailored to the anatomical and physiological characteristics of elementary school players, formulates mathematical calculation models for 2D side-view video, establishes automated detection rules for typical form errors, and structures an automated feedback and drill system linked with scoring.

Baseline Swing Mechanics and Phase-by-Phase Kinematic Targets

Although a batting swing is a continuous movement, in 2D side-view video analysis, dividing the motion into 5 distinct time-series phases is effective for evaluating positional relationships (alignment) and joint angles of body parts.

1. Setup / Stance

The goal of the setup phase is to eliminate unnecessary muscular tension and establish a power position to store mechanical energy. The stance width is set to 1.2 to 1.5 times the torso length (the distance between the shoulder and hip joints), which is slightly wider than shoulder width, with weight distributed on the balls of both feet. Knees are lightly pinched inward rather than flared outward, maintaining tension in the inner thighs and increasing lower-body rigidity.

Slight hip flexion establishes a torso forward tilt of approximately $25^\circ \sim 35^\circ$ relative to the vertical line, activating the hamstrings and gluteal muscles. The grip is positioned at ear-to-shoulder height (Y -axis) and aligned with the vertical line of the rear foot's toes to heel (X -axis), preventing casting and ensuring a level-to-slightly-downward initial movement angle.

2. Loading / Stride Phase

This phase involves simultaneous linear motion toward the pitcher and rotary power storage (loading) into the rear hip. As the lead foot lifts, weight shifts into the rear hip joint. It is crucial that the rear knee does not sway outward beyond the lateral boundary of the rear foot

(prevention of excessive negative weight shift).
While the lead foot strides toward the pitcher, the grip and upper body remain loaded toward the rear, stretching the lead scapula (Scapular Load). This opposing motion creates hip-shoulder separation. During the stride, horizontal movement of the head (eye level) is kept to a minimum, constrained to a parallel translation of the body axis.

3. Foot Strike / Foot Plant

Explosive rotational motion initiates the moment the lead heel plants on the ground (Heel Plant). Flexion of the lead knee joint stops upon heel plant, forming a firm "wall" (braking). This ground reaction force (approximately 123% or more of body weight) drives momentum transfer from the lower body to the upper body.
The kinetic energy follows a strict kinematic sequence: the pelvis (Hip) reaches peak rotational velocity first (approx. 714 deg/s), followed sequentially by the thorax/shoulders (Shoulder, approx. 937 deg/s), arms/hands (Hands), and bat (Bat, approx. 31 m/s). At rotation initiation, the angle between the lead forearm and torso (Early Connection) is maintained within $80^\circ \sim 105^\circ$, aligning the bat onto the rotational plane of the torso.

4. Impact Phase

This phase delivers stored rotational energy to the ball with maximum efficiency. At impact, seven key checkpoints should be satisfied: lead palm facing down and rear palm facing up (Top hand palm-up, Bottom hand palm-down), rear foot on toe, extended front leg wall, rear elbow slotted near the torso, head positioned inside the triangle formed by both knees, line-of-sight fixed on the contact point, and structural alignment between head, rear knee, and ground.
While maintaining the torso forward tilt established during setup, the optimal bat vertical angle (Attack Angle) at impact is $+5^\circ \sim +15^\circ$ (a slight upper trajectory) to align with the ball's incoming flight path. An attack angle exceeding $+20^\circ$ reduces exit velocity and causes pop-ups.

5. Follow-through Phase

This deceleration phase stabilizes the swing path without abruptly killing inertial forces. Rather than forcibly rolling the wrists, the bat head should be extended broadly toward center field. Maintaining the palm-up/palm-down relationship as long as possible ensures an extended contact zone. Maintaining the torso tilt and head position established at impact allows the batter to finish in balance between both feet.

Swing Phase	**Primary Target Joints / Body Segments**	**Baseline Parameters / Kinematic Targets**	**Physiological & Biomechanical Functions**
Setup	Both ankles, Hips, Shoulders, Grip	Stance width ratio 1.2–1.5x, Torso tilt	Lower-body stability, glute/hamstring

		$25^\circ \sim 35^\circ$, Grip height at ear–shoulder level	activation, casting prevention
Stride	Rear knee, Rear hip, Head, Grip	Rear knee inside rear ankle boundary, Minimal head displacement	Energy dissipation prevention, Visualizing hip-shoulder stretch
Foot Strike	Lead knee, Pelvis angle, Shoulder angle, Forearm-Torso angle	Lead knee angle fixed/extended, Early Connection $80^\circ \sim 105^\circ$	Ground reaction force capture, Kinematic sequence initiation
Impact	Bat angle, Both elbows, Lead leg, Torso angle	Attack angle $+5^\circ \sim +15^\circ$, Maintained torso tilt, Rear elbow slotted	Pitch trajectory matching, Maximum energy transfer
Follow-through	Wrist joints, Center of gravity, Torso tilt	Elimination of early wrist roll-over, Maintained torso inclination	Extension of contact zone, Swing path stabilization

2D Side-View Pose Estimation & Kinematic Metric Mathematical Models

Coordinate series obtained from a single 2D side-view camera (e.g., smartphone) are affected by camera-to-subject distance, body scale, and individual anatomical differences. Therefore, raw 2D pixel coordinates (x, y) must be converted into scale-invariant angular metrics and relative coordinate systems normalized by torso length.

1. 2D Vector Angle Calculation Logic

By defining vectors connecting keypoints, joint angles and segment inclinations are calculated using vector dot products and the two-argument arctangent function (arctan2). The inclination angle θ of a vector connecting keypoint $A(x_a, y_a)$ and keypoint $B(x_b, y_b)$ relative to the horizontal axis is defined as:

$$\theta = \text{arctan2}(y_b - y_a, x_b - x_a)$$

The joint flexion angle ϕ formed by three points A, B, C (e.g., shoulder, elbow, wrist) is derived from the dot product of vectors BA and BC:

$$\phi = \arccos\left(\frac{BA \cdot BC}{\|BA\|\|BC\|}\right)$$

2. Primary Kinematic Metrics for 2D Side View

- **Normalized Stance Width (W_{stance}):**
  Horizontal X-axis distance between lead ankle and rear ankle $|X_{lead\_ankle} - X_{rear\_ankle}|$ normalized by torso length L_{torso} (distance between shoulder and hip joint keypoints):

  $$W_{stance} = \frac{|X_{lead\_ankle} - X_{rear\_ankle}|}{L_{torso}}$$

  A target range of $1.2 \leq W_{stance} \leq 1.5$ during setup and foot strike balances lower-body stability and rotational ease.
- **Torso Forward Tilt Angle (θ_{torso}):**
  Angle between the torso vector V_{torso} (hip to shoulder) and the vertical reference vector $V_{vertical}$:

$$\theta_{torso} = \arccos\left(\frac{V_{torso} \cdot V_{vertical}}{\|V_{torso}\|}\right)$$

Maintaining θ_{torso} within $25^\circ \sim 35^\circ$ from setup to impact with a variance under 10° serves as the benchmark for posture preservation.

- **Grip Loading Vector (D_{grip_x}, H_{grip_y}):**

  Grip (wrists) positional coordinates relative to rear ankle X -coordinate and shoulder/ear Y -coordinates. Target H_{grip_y} sits between ear and shoulder height, and horizontal offset D_{grip_x} stays within the rear foot support boundary to prevent casting.

- **Early Connection Angle (θ_{EC}):**

  Angle between torso vector V_{torso} and lead forearm vector V_{lead_arm}. Evaluated at swing initiation (rotation start frame), with an optimal range of $80^\circ \leq \theta_{EC} \leq 105^\circ$.

- **Lead Knee Blocking Index ($\Delta\phi_{knee}$):**

  Difference between lead knee flexion angle at foot strike $\phi_{knee}(t_{fs})$ and at impact $\phi_{knee}(t_{imp})$:

$$\Delta\phi_{knee} = \phi_{knee}(t_{imp}) - \phi_{knee}(t_{fs})$$

  $\Delta\phi_{knee} \geq 0$ (knee angle maintained or extended) indicates proper blocking, whereas $\Delta\phi_{knee} < 0$ (further collapse/flexion) indicates front knee collapse.

- **Head Translation Ratio (R_{head}):**

  Horizontal X -axis displacement of the head keypoint from setup X_{head_setup} to impact X_{head_impact}, normalized by torso length L_{torso}:

$$R_{head} = \frac{X_{head\_impact} - X_{head\_setup}}{L_{torso}}$$

An $R_{head} > 0.25$ indicates excessive forward lunging/axis drift.

- **Estimated Attack Angle (θ_{AA}):**
Tangential angle of the quadratic curve fitted to wrist/bat trajectory across frames surrounding impact. 0° represents horizontal, positive angles represent an upward trajectory, and negative angles represent downward trajectory.

Metric Name	**Target Phase**	**Keypoints Used**	**Mathematical Calculation / Vector Definition**	**Baseline Target Range**
Normalized Stance Width	Setup / Foot Strike	Both ankles, Both hips, Both shoulders	Ankle X-distance $\div L_{torso}$	$1.2 \sim 1.5$
Torso Forward Tilt	Setup to Impact	Both hips, Both shoulders	Angle θ_{torso} between torso vector and vertical	$25^\circ \sim 35^\circ$ (variance $\leq 10^\circ$)
Grip Loading Vector	Setup / Loading	Wrists, Rear ankle, Shoulder, Ear	Relative coordinates to rear ankle X and shoulder/ear Y	Height: Ear–Shoulder / Depth: Over rear foot
Early Connection Angle	Foot Strike (Rotation Start)	Lead shoulder, Lead wrist, Rear hip	Angle θ_{EC} between lead arm vector and torso axis	$80^\circ \sim 105^\circ$

Lead Knee Blocking	Foot Strike to Impact	Lead hip, Lead knee, Lead ankle	Impact knee angle minus foot strike knee angle	$\Delta\phi_{knee} \geq 0^\circ$ (extension/holding)
Head Vector Displacement	Setup to Impact	Head (Ear/Nose), Rear ankle, Torso length	Head horizontal movement $\div L_{torso}$	≤ 0.25 (Axis stability)
Estimated Attack Angle	Impact	Wrists (Grip), Bat tip	Trajectory tangent angle around impact	$+5^\circ \sim +15^\circ$ (Slight upper)

Common Swing Errors in Youth Baseball and Automated Detection Rules

Batting errors in young players frequently stem from undeveloped muscular strength, oversized or heavy bats, and misinterpretation of coaching cues (e.g., "chop down on the ball" or "don't open up"). The rule-based diagnostic logic for detecting these mechanical errors from 2D side-view video is detailed below.

1. Door Swing / Casting

When a player attempts to swing with an upright torso lacking forward tilt, centrifugal force pulls the grip and bat away from the torso into an outer-loop trajectory. Common when relying solely on arm strength or using an overly heavy bat. Automated detection rules flag this if Early Connection Angle $\theta_{EC} > 105^\circ$ at swing initiation, or if the horizontal distance between wrists and chest $D_{wrist-chest}$ relative to torso length exceeds $D_{wrist-chest} / L_{torso} > 0.6$ during mid-swing.

2. Forward Axis Drift / Rushing

Occurs when a player fails to load into the rear hip joint during stride and lunges forward toward the pitcher prior to foot plant. This breaks the kinematic sequence and prevents lower-body rotational power from transferring to the upper body. Automated detection flags this if the rear knee X -coordinate sways past the rear ankle toward the catcher during backswing (excessive negative weight shift), or if head translation at impact exceeds

$R_{head} > 0.25$.

3. Arms-Only Swing / One-Piece Swing

Occurs when the lower body (pelvis) and upper body (thorax) fail to create hip-shoulder separation, rotating simultaneously as a single unit. This negates the stretch-shortening cycle of the core and significantly decreases swing speed. Automated detection rules flag this if there is no rotational phase lag between pelvic and shoulder vectors at foot strike, or if minimal elbow flexion angle change occurs throughout the swing.

4. Excessive Upper Swing / Early Extension

Driven by an overemphasis on lifting the ball or an inability to maintain setup forward tilt, causing the pelvis to thrust forward while the torso extends upright prior to impact (Early Extension). The bat head drops, producing an excessively steep positive attack angle. Automated detection flags this if calculated attack angle $\theta_{AA} > +20^\circ$ or if torso forward tilt θ_{torso} decreases by more than 10° (moving upright) at impact compared to setup.

5. Collapsed Lead Side

Occurs when the front knee flares open or continues to flex deeply after foot strike, failing to block forward momentum. Power leaks forward and the rotational axis collapses. Automated detection flags this if lead knee angle change $\Delta\phi_{knee} < -5^\circ$ (indicating increased flexion after foot plant) or if lead knee X -coordinate drifts forward past lead ankle X -coordinate.

Swing Error Pattern	Root Biomechanical Cause	2D Side-View Detection Rules (Mathematical Conditions)	Impact on Swing & Ball Flight
Door Swing / Casting	Upright posture centrifugal detachment, Overweight bat	$\theta_{EC} > 105^\circ$ or $D_{wrist-chest} / L_{torso}$	Casting trajectory, late contact, weak contact
Forward Axis Drift / Rushing	Insufficient rear hip loading, Braking failure	Impact $R_{head} > 0.25$ or $X_{head} > X_{lead\_knee}$	Decreased contact consistency, poor off-speed adjustment
Arms-Only Swing	Lack of	Pelvis-shoulder	Reduced bat speed,

	hip-shoulder separation, Core instability	rotation phase lag ≈ 0, No elbow angle delta	diminished exit velocity
Excessive Upper Swing	Early loss of torso tilt (Early Extension)	$\theta_{AA} > +20^\circ$ and loss of torso tilt (Upright drift)	High pop-up rate, under-ball skimming contact
Collapsed Lead Side	Weak lead leg blocking, Insufficient ground reaction	$\Delta\phi_{knee} < -5^\circ$ or lead knee forward drift	Attenuated rotational acceleration, off-center contact

Scoring Algorithm and Automated Pipeline Design

Overall score calculation (100-point scale) utilizes a weighted deduction method based on the biomechanical importance of each swing phase. The automated pipeline ingests raw video, extracts 2D skeletal keypoint series using pose estimation models (e.g., YOLOv8-pose, MediaPipe, HRNet), and aligns time-series frames into baseline swing phases using Dynamic Time Warping (DTW) or keypose detection models. Once aligned, geometric angles and vectors are computed and passed to the scoring engine and feedback generator.

1. Phase Weightings and Penalty Function

Total Score S_{total} is defined by:

$$S_{total} = 100 - \sum_{k=1}^{N} W_k \cdot P_k(M_k, M_{ref,k})$$

Where W_k represents weight coefficients for metric k ($\sum W_k = 100$), M_k is the measured metric value, $M_{ref,k}$ is the target baseline range, and P_k is a regularized penalty function ($0.0 \sim 1.0$) proportional to deviation from target thresholds.
Phase weight allocations are assigned based on kinematic contribution:

- **Setup Phase: Weight 10%** (Metrics: Stance width ratio W_{stance}, Torso tilt θ_{torso}, Initial grip position)

- **Stride Phase: Weight 20%** (Metrics: Rear knee sway, Head translation control)
- **Foot Strike Phase: Weight 25%** (Metrics: Early Connection Angle θ_{EC}, Hip-shoulder separation lag)
- **Impact Phase: Weight 35%** (Metrics: Attack angle θ_{AA}, Lead knee blocking $\Delta\phi_{knee}$, Torso tilt preservation, Final head position)
- **Follow-through Phase: Weight 10%** (Metrics: Posture tilt maintenance, Finish balance)

2. Feedback Generation Logic

The scoring engine identifies the primary error yielding the highest point deduction and outputs a 3-tier diagnostic report:

1. **Calculated Scores** (Phase-specific and overall score out of 100).
2. **Error Point Explanation** (Specific body alignment faults and biomechanical factors).
3. **Remedial Action Plan** (Targeted drill assignments and actionable verbal cues).

Error-Specific Improvement Protocols and Remedial Drills

To correct detected form errors, vague commands (e.g., "don't open up" or "swing harder") are replaced with dedicated corrective drills and clear verbal cues designed for youth players.

1. Door Swing Correction Protocol

Forcing players to "keep elbows tucked" restricts wrist freedom and causes muscular rigidity. Instead, maintaining proper torso tilt and feeling the bat path emerge naturally inside the rotational axis (inside-out path) is emphasized.

- **Cross-Chest Rotation Drill**: Perform rotations without a bat while crossing both arms over the chest in setup posture. Teaches the relationship between torso tilt and rotational axis, preventing arm casting.
- **Inside-Out Tee Drill**: Place a tee on the inside lower corner and practice hitting balls toward the opposite field (right field for right-handed batters). Forces grip-first inside-out bat path.

2. Forward Axis Drift Correction Protocol

Develops lower-body loading into the rear hip crease (groin area) to lock the pelvis and delay upper body rotation until foot plant.

- **5-Second Rear Leg Hold Drill**: Lift the lead leg high and hold for 5 seconds before striding. Builds hip loading awareness, single-leg balance, and core stability.
- **Single-Leg Swing Drill**: Execute swings while remaining balanced on the rear leg to eliminate excessive forward translation of the body axis.

3. Arms-Only Swing Correction Protocol

Teaches the time-lag sequence where lower-body rotation initiates first, followed by upper-body turn.

- **Hugged-Bat Lower-Body Drill**: Hold a bat horizontally across the chest, rendering arms inactive, and hit balls off a tee using only lower-body rotation. Connects lower-body ground drive directly to core turn.

4. Excessive Upper Swing Correction Protocol

Excessive upward attack angles ($> +20^\circ$) stem from rear shoulder dropping and early loss of torso forward tilt (Early Extension).

- **High-Grip Stop Drill**: Take a wide stance without striding. Freeze the swing immediately after impact, ensuring the grip finishes higher than the bat head. Promotes level-to-slightly-elevated contact control.

Diagnostic Error	Automated Phenomenon Explanation	Root Biomechanical Cause	Assigned Remedial Drill	Instructional Verbal Cue
Door Swing	Bat is casting outward away from the body	Upright posture, lack of inside-out trajectory	Cross-Chest Rotation Drill, Inside-Out Tee Drill	"Keep your bow posture and turn around your spine."
Forward Axis Drift	Body lunges forward, causing unstable contact point	Poor rear hip loading, lead leg braking failure	5-Second Rear Leg Hold Drill, Single-Leg Swing Drill	"Sit into your back hip and hold for 5 seconds."
Arms-Only Swing	Upper and lower body rotate together without sequence	Lack of pelvic-thoracic separation	Hugged-Bat Lower-Body Drill, Stationary Tee Work	"Turn your hips first, let your arms follow."
Excessive Upper Swing	Bat cuts under the ball, leading to weak pop-ups	Early Extension, dropping rear shoulder	High-Grip Stop Drill, Hoop Rotation Drill	"Keep your hands higher than your head after contact."
Collapsed Lead Side	Front knee collapses at impact, causing axis	Front leg strength deficit, ground force failure	Front Leg Stiff-Stop Drill, Single-Leg Swing Drill	"Push the ground back firmly with your front

	wobble			foot."

Integrating these kinematic parameters, mathematical detection rules, and remedial drill mappings into the core system architecture enables high-precision automated analysis and actionable coaching feedback from standard 2D side-view video captured on mobile devices.